

Session 3 — Analysis I

Solutions

These solutions are written to be read on their own. Each exercise is answered twice: once by computation, with every hypothesis stated where it does work, and once by a **picture**. For the differential equations the picture is the primary answer — the flow, the equilibria and the stability are all visible before any algebra — and the computation is there to confirm it and to make it exact.

Two habits are used throughout and are worth adopting. **Substitute to check**: a candidate solution is verified in one line, which is a complete proof. And **state the interval**: a solution is a function together with an interval, and for nonlinear equations that interval can be smaller than $\mathbb{R}$.

Exercise 1 Differentiate $t^2 - 4t + 7$, $5e^{-2t}$, e^{-t^2} ; study te^{-t} ; diagnose $\frac{d}{dt}e^{-t/2} = (-t/2)e^{(-t/2)-1}$.

What this exercise uses. Three rules. *Power*: $(t^n)' = nt^{n-1}$, valid when the variable is in the **base**. *Exponential*: $(e^u)' = u'e^u$, valid when the variable is in the **exponent**. *Product*: $(fg)' = f'g + fg'$.

Solution.

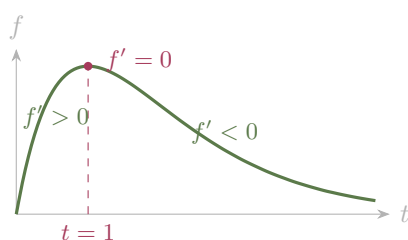
(a) $(t^2 - 4t + 7)' = 2t - 4$ (power rule, term by term). $(5e^{-2t})' = 5 \cdot (-2)e^{-2t} = -10e^{-2t}$ (chain rule: outside $e^u \rightarrow e^u$, inside $u = -2t \rightarrow -2$). $(e^{-t^2})' = -2te^{-t^2}$ (same, with $u = -t^2$, $u' = -2t$).

(b) $f(t) = te^{-t}$ gives, by the product rule,

$$f'(t) = 1 \cdot e^{-t} + t \cdot (-e^{-t}) = (1 - t)e^{-t}.$$

Since $e^{-t} > 0$ for all t , the sign of f' is the sign of $1 - t$: positive for $t < 1$, negative for $t > 1$. So f increases then decreases and $t = 1$ is a **maximum**, with $f(1) = e^{-1} \approx 0.37$. Also $f(0) = 0$ and $f(t) \rightarrow 0$ as $t \rightarrow +\infty$.

(c) They applied the **power rule** to an exponential. It does not apply because the variable sits in the exponent, not in the base; $e^{-t/2}$ is not of the form t^n . The correct derivative is $-\frac{1}{2}e^{-t/2}$. The question that prevents the error: *is the variable in the base or in the exponent?*



$f(t) = te^{-t}$: rise, one maximum, decay.

What the picture shows. The maximum is not found by evaluating f anywhere — it is found by reading the *sign* of f' on either side of the point where it vanishes. This is the same argument that will decide stability in Exercise 10.

Common mistake. Writing $(fg)' = f'g'$. Test it on $f = g = t$: it would give 1, whereas $(t^2)' = 2t$.

Exercise 2 Compute $\int_0^2 (3t^2 - 4t + 1) dt$ and $\int \frac{2t}{1+t^2} dt$; compute $\int_0^T e^{-t/\tau} dt$ and its limit; explain the condition $n \neq -1$.

What this exercise uses. The fundamental theorem, *with its hypothesis*: if f is **continuous** on an interval I and $G' = f$ on I , then $\int_a^b f = G(b) - G(a)$. Two antiderivatives differ by a constant **on an interval** — the word matters, see (c).

Solution.

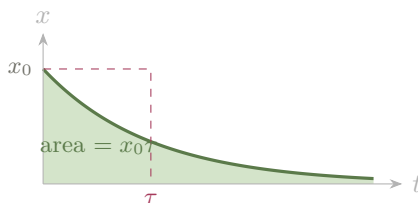
(a) $\int_0^2 (3t^2 - 4t + 1) dt = [t^3 - 2t^2 + t]_0^2 = 8 - 8 + 2 = 2.$

For the second, substitute $u = 1 + t^2$, $du = 2t dt$: $\int \frac{du}{u} = \ln |u| + C = \ln(1 + t^2) + C$ (the absolute value is harmless since $1 + t^2 > 0$). *Check:* $\frac{d}{dt} \ln(1 + t^2) = \frac{2t}{1+t^2}$ ✓

(b) $\int_0^T e^{-t/\tau} dt = [-\tau e^{-t/\tau}]_0^T = \tau(1 - e^{-T/\tau}) \xrightarrow{T \rightarrow \infty} \tau.$

So the total area under a decaying exponential is **exactly its time constant**: τ is not only a clock, it is also an area.

(c) Because $\frac{t^{n+1}}{n+1}$ divides by $n+1$, which vanishes at $n = -1$. The gap is filled by the logarithm: $\int t^{-1} dt = \ln |t| + C$ on any interval not containing 0. On $\mathbb{R}^*$, which is *not* an interval, two antiderivatives of $1/t$ need not differ by a single constant — the constant may differ on the two sides.



The shaded area equals the area of the dashed rectangle of width τ .

What the picture shows. The infinite tail contributes exactly as much as a rectangle of height x_0 and width τ . That is a useful mental image: a decaying exponential “is worth” τ units of time at full strength.

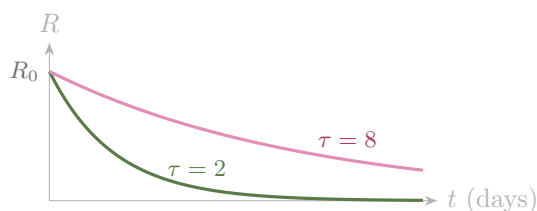
Exercise 3 Write $e^{-t/5}$ as e^{at} ; show $R' = -R/\tau$; compare $\tau = 2$ and $\tau = 8$.

What this exercise uses. e^{at} decays iff $a < 0$, and for $a < 0$ one writes $a = -1/\tau$ with $\tau = 1/|a| > 0$.

Solution.

(a) $e^{-t/5} = e^{at}$ with $a = -\frac{1}{5}$. Decay occurs precisely when $a < 0$ (for $a > 0$ the function grows, for $a = 0$ it is constant).

- (b) $R(t) = R_0 e^{-t/\tau}$ gives $R'(t) = -\frac{1}{\tau} R_0 e^{-t/\tau} = -\frac{1}{\tau} R(t)$. In words: *the rate of forgetting is proportional to how much is still remembered* — and the constant of proportionality is $1/\tau$.
- (c) Nothing needs computing. Rearranging (b), $|R'|/R = 1/\tau$: the **relative** rate of loss is $1/\tau$. A smaller τ therefore means a larger relative rate, so $\tau = 2$ forgets faster.



Same R_0 , same law, two time constants.

What the picture shows. Both curves start together and neither ever reaches zero. What τ controls is the *steepness at every height*, not a finishing time.

Common mistake. Reading τ as “the time the memory lasts”. At $t = 2\tau$ the curve is still clearly positive, and it stays positive for ever.

Exercise 4 Classify as integration or as an equation involving the unknown, then solve with $x(0) = 1$: $\dot{x} = 3t^2$, $\dot{x} = e^{-t}$, $\dot{x} = -2x$, $\dot{x} = -2x + 5$.

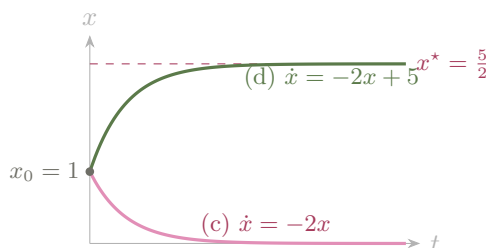
What this exercise uses. If the right-hand side does not mention x , integrate: the solution is $x(t) = x_0 + \int_0^t f(s)ds$. If it does mention x , integration is unavailable — you would need $x(t)$ to perform it — and for $\dot{x} = ax + b$ ($a \neq 0$) the solution is $x^* + (x_0 - x^*)e^{at}$ with $x^* = -b/a$.

Solution.

- (a) **Integration.** $x(t) = t^3 + 1$, on $\mathbb{R}$.
- (b) **Integration.** $x(t) = 1 + \int_0^t e^{-s} ds = 2 - e^{-t}$, on $\mathbb{R}$. Note $x(t) \rightarrow 2$: an integration problem can also have a horizontal asymptote.
- (c) **Equation in the unknown.** $a = -2$, so $x(t) = e^{-2t}$ on $\mathbb{R}$.
- (d) **Equation in the unknown.** $x^* = -5/(-2) = \frac{5}{2}$, so

$$x(t) = \frac{5}{2} + \left(1 - \frac{5}{2}\right)e^{-2t} = \frac{5}{2} - \frac{3}{2}e^{-2t}, \quad t \in \mathbb{R}.$$

Check: $\dot{x} = 3e^{-2t}$, and $-2x + 5 = -5 + 3e^{-2t} + 5 = 3e^{-2t}$ ✓



Same initial value, same rate constant, different destinations.

What the picture shows. (c) and (d) differ only by the constant b , and the constant only moves the **destination**: one closes the gap to 0, the other the gap to $\frac{5}{2}$, and both at the same speed $\tau = \frac{1}{2}$.

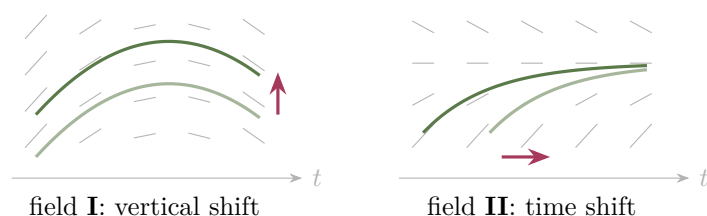
Common mistake. Answering (d) with $e^{-2t} + 5$ — an antiderivative habit. Substituting it gives $-2e^{-2t}$ on the left and $-2e^{-2t} - 5$ on the right: not equal.

Exercise 5 For the two printed slope fields: does the slope depend on t , on x , or both? Which is $\dot{x} = f(t)$ and which is $\dot{x} = f(x)$? Sketch two solutions in each and say how they are related.

What this exercise uses. A solution is a curve tangent to the field at each of its points. Two invariances: if $\dot{x} = f(t)$ and x is a solution, so is $x + c$ (vertical shift); if $\dot{x} = f(x)$ and x is a solution, so is $t \mapsto x(t + c)$ (time shift).

Solution.

- (a) In field **I** the segments are parallel along each **vertical** line: the slope depends on t alone. In field **II** they are parallel along each **horizontal** line: the slope depends on x alone.
- (b) Field I is $\dot{x} = f(t)$; field II is $\dot{x} = f(x)$ (autonomous).
- (c) In field I the two curves are **vertical translates** of each other: the proof is one line, since $\frac{d}{dt}(x + c) = \dot{x} = f(t)$, the right-hand side being unaffected by the shift. In field II the upper curve is the lower one **shifted in time**, $y(t) = x(t + c)$; here $\dot{y}(t) = \dot{x}(t + c) = f(x(t + c)) = f(y(t))$.



What the picture shows. The two invariances are visible before any computation: in I you may slide a solution *up*, in II you may slide it *sideways*.

Remark. The time-shift statement is about producing a solution from a solution. It does not say that any two solutions of an autonomous equation are time shifts of one another: a curve lying below an equilibrium is never a shift of one lying above it, since no solution crosses an equilibrium (Exercise 8).

Exercise 6 Someone proposes $\varphi(x) = -x^2/2 + 3x$ as the solution of $\dot{x} = -(x - 3)$.

What this exercise uses. The unknown of $\dot{x} = f(x)$ is a function of t . Antidifferentiating f produces a function of x . The two are objects of different types.

Solution.

(a) φ is a function of x ; a solution must be a function of t . The types do not match, so φ cannot be a solution — no computation required.

(b) If x is any differentiable function of t , the chain rule gives

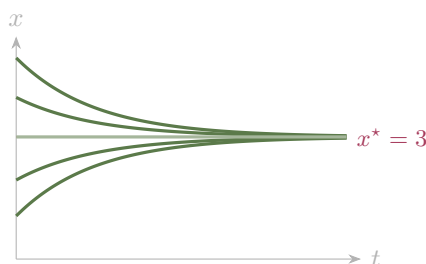
$$\frac{d}{dt} \varphi(x(t)) = \varphi'(x(t)) \dot{x}(t) = (3 - x) \dot{x}.$$

For this to equal the prescribed rate $-(x - 3) = 3 - x$ one would need $(3 - x)\dot{x} = 3 - x$, i.e. $\dot{x} = 1$ wherever $x \neq 3$ — a different equation from the one asked.

(c) The equation is $\dot{x} = ax + b$ with $a = -1$, $b = 3$, so $x^* = 3$ and

$$x(t) = 3 + (x_0 - 3)e^{-t}, \quad t \in \mathbb{R}.$$

Check: $\dot{x} = -(x_0 - 3)e^{-t}$, while $-(x - 3) = -(3 + (x_0 - 3)e^{-t} - 3) = -(x_0 - 3)e^{-t}$ ✓



Solutions of $\dot{x} = -(x - 3)$ from four initial values.

What the picture shows. Every curve approaches the constant solution $x \equiv 3$ and none of them touches it. The gap $x - 3$ is what decays; the quantity itself decays only if x^* happens to be 0.

Common mistake. Believing the objection is about the arithmetic. It is about the type of the object produced — which is why the type argument settles it faster than any computation.

Exercise 7 $\dot{x} = ax + b$: verify the formula; analyse $\dot{x} = -x + 4$ and $\dot{x} = 3x - 6$; and the case $a = 0$.

What this exercise uses. $x^* = -b/a$ is the value at which the rate vanishes. Writing $u = x - x^*$ gives $\dot{u} = au$: the gap to the equilibrium obeys the homogeneous equation, so it decays iff $a < 0$.

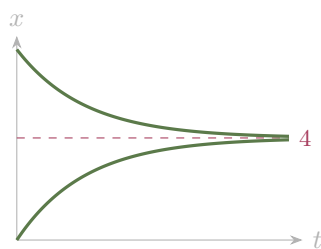
Solution.

(a) With $x(t) = x^* + (x_0 - x^*)e^{at}$:

$$\dot{x} = a(x_0 - x^*)e^{at}, \quad ax + b = \underbrace{ax^* + b}_{=0} + a(x_0 - x^*)e^{at},$$

so both sides agree for every t , and $x(0) = x_0$. Valid on all of $\mathbb{R}$: the solution of a linear equation is global.

- (b) $x^* = 4$, $a = -1$ so $\tau = 1$, and $a < 0$: **stable**. From $x_0 = 10$ the solution decreases towards 4; from $x_0 = 0$ it increases towards 4. Same curve, travelled in opposite directions, and neither reaches 4 in finite time.
- (c) $x^* = 2$, $a = 3 > 0$: **unstable**. From $x_0 = 1.9$ the gap $x - 2 = -0.1e^{3t}$ grows in magnitude, so $x \rightarrow -\infty$; from $x_0 = 2.1$, $x \rightarrow +\infty$. A displacement of one tenth is enough to decide the fate.
- (d) $x^* = -b/a$ is undefined: there is no equilibrium unless $b = 0$ as well. Directly, $\dot{x} = b$ is an integration problem with $x(t) = x_0 + bt$ — linear growth for ever, no destination.



(b) $a < 0$: **stable**



(c) $a > 0$: **unstable**

What the picture shows. The same formula produces both pictures; only the sign of a differs. On the left the two curves are squeezed together, on the right they are prised apart — and that is the whole content of the word *stability*.

Exercise 8 $\dot{x} = 2\sqrt{|x|}$: uniqueness at $x_0 = 1$ and at $x_0 = 0$; and why the solution of $\dot{x} = -x+4$ from 10 never reaches 4.

What this exercise uses. Cauchy–Lipschitz: f continuous and **locally Lipschitz in x** gives a unique maximal solution through each point. A C^1 function is locally Lipschitz. Uniqueness has a geometric form: two distinct integral curves never meet.

Solution.

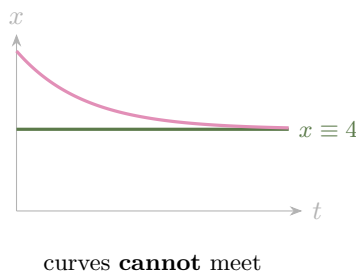
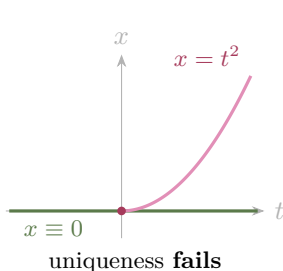
- (a) **Yes.** Near $x = 1$ we have $|x| = x$ and $f(x) = 2\sqrt{x}$, which is C^1 on a neighbourhood of 1 (its derivative $1/\sqrt{x}$ is continuous there), so f is locally Lipschitz at $x = 1$ and the theorem applies.
- (b) Both are solutions. For $x \equiv 0$: $\dot{x} = 0$ and $2\sqrt{|0|} = 0$ ✓ For $x(t) = t^2$ with $t \geq 0$: $\dot{x} = 2t$ and $2\sqrt{|t^2|} = 2|t| = 2t$ ✓ Both satisfy $x(0) = 0$. The hypothesis that fails is the Lipschitz condition, **at the single point** $x = 0$: for $h > 0$,

$$\frac{|f(h) - f(0)|}{h} = \frac{2\sqrt{h}}{h} = \frac{2}{\sqrt{h}} \rightarrow +\infty,$$

so no constant L can bound the ratio near 0. (One can even glue: stay at 0 until any time $t_1 > 0$, then leave along $(t - t_1)^2$ — infinitely many solutions.)

- (c) Here $f(x) = -x + 4$ is C^1 on $\mathbb{R}$, so solutions are unique. The constant function $x \equiv 4$ is a solution. If the solution starting at 10 took the value 4 at some time t_1 , then two distinct

solutions would pass through the point $(t_1, 4)$, contradicting uniqueness. Hence it approaches 4 without ever attaining it — which the formula confirms, since $e^{-t} > 0$ for all t .



What the picture shows. On the left the hypothesis fails at one point and two curves leave it. On the right the hypothesis holds, so the constant solution acts as a **barrier** that the other solution may approach but never cross.

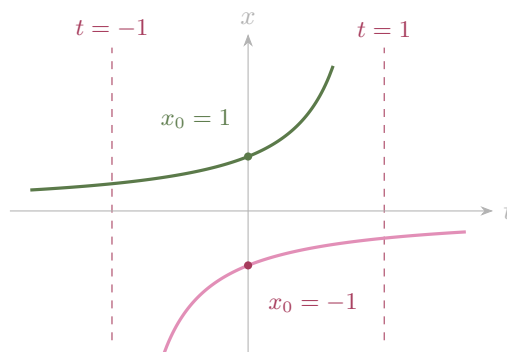
Remark. This is why, in dimension one, equilibria separate the line into regions a trajectory can never leave — and hence why the phase line of Exercise 10 tells the whole story.

Exercise 9 Maximal solutions of $\dot{x} = x^2$ through $x_0 = 1$ and $x_0 = -1$; and why smoothness of f does not prevent blow-up.

What this exercise uses. A solution is a function **together with an interval**. It is *maximal* when it admits no extension; under the Cauchy–Lipschitz hypotheses the maximal interval is open, since at an included endpoint the theorem could be applied again and the solution continued.

Solution.

- (a) $x(t) = \frac{1}{1-t}$ gives $\dot{x} = \frac{1}{(1-t)^2} = x^2$ ✓ and $x(0) = 1$ ✓ It is a solution on $(-\infty, 1)$, and that interval is maximal: $|x(t)| \rightarrow +\infty$ as $t \rightarrow 1^-$, so no extension past 1 exists.
The answer is not $\mathbb{R} \setminus \{1\}$ because that set is **not an interval**. The formula also defines a solution on $(1, +\infty)$, but it is a *different* solution — it does not pass through $(0, 1)$, and no solution connects the two branches.
- (b) $x(t) = \frac{-1}{1+t}$ gives $\dot{x} = \frac{1}{(1+t)^2} = x^2$ ✓ and $x(0) = -1$ ✓ Here the blow-up is in the *past*: the maximal interval is $(-1, +\infty)$, and forwards in time the solution exists for ever, increasing towards 0.
- (c) Smoothness of f is a **local** hypothesis and yields a **local** conclusion: existence and uniqueness near each point. It says nothing about how long the solution can be continued. Here the growth of f is what does the damage: the larger x is, the faster it grows, fast enough to reach infinity in finite time.



Same equation $\dot{x} = x^2$, same smooth f , two different maximal intervals.

What the picture shows. The vertical dashed lines are not features of the equation — they are features of the *solutions*. Which is exactly why the interval is part of the answer.

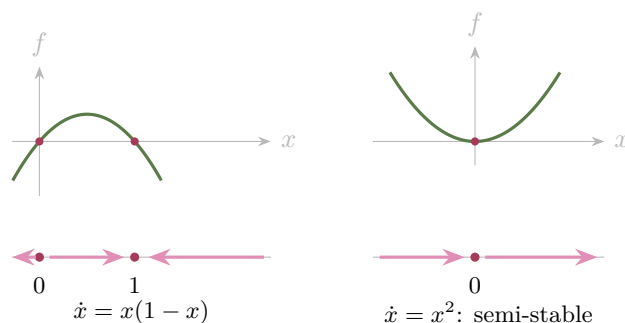
Common mistake. Writing “the solution is $1/(1-t)$ ” and stopping. Without its interval the statement is incomplete, and at $t = 2$ the formula returns -1 , a number that belongs to another solution entirely.

Exercise 10 Phase lines for $\dot{x} = -x + 3$ and $\dot{x} = x - 3$; equilibria and stability of $\dot{x} = x(1-x)$; and $\dot{x} = x^2$.

What this exercise uses. For $\dot{x} = f(x)$: equilibria are the zeros of f ; the flow moves right where $f > 0$ and left where $f < 0$. If f is C^1 and $f(x^*) = 0$, then $f'(x^*) < 0$ gives asymptotic stability and $f'(x^*) > 0$ instability; if $f'(x^*) = 0$ the criterion is silent. The reason is that the displacement $\delta = x - x^*$ obeys $\dot{\delta} = f'(x^*)\delta + o(\delta)$.

Solution.

- (a) Both have $x^* = 3$. For $\dot{x} = -x + 3$: $f' = -1 < 0$, arrows point inwards from both sides, **stable**. For $\dot{x} = x - 3$: $f' = +1 > 0$, arrows point outwards, **unstable**. Neither equation had to be solved.
- (b) $f(x) = x(1-x)$ vanishes at $x = 0$ and $x = 1$; $f'(x) = 1 - 2x$, so $f'(0) = 1 > 0$ (**unstable**) and $f'(1) = -1 < 0$ (**stable**). From $x_0 = 0.1$ the solution increases to 1. From $x_0 = -0.1$ it decreases without bound — and in fact reaches $-\infty$ in finite time, since for $x < 0$ we have $\dot{x} = x - x^2 \leq -x^2$, and $\dot{x} = -x^2$ already blows up in finite time.
- (c) $f(x) = x^2$ has $x^* = 0$ and $f'(0) = 0$: the criterion says nothing. But $f \geq 0$ everywhere, so the flow points **right on both sides**: the equilibrium attracts from the left and repels on the right. Such a point is called **semi-stable** — and it is neither stable nor of the unstable type described in (a), which is precisely why the theorem declines to answer.



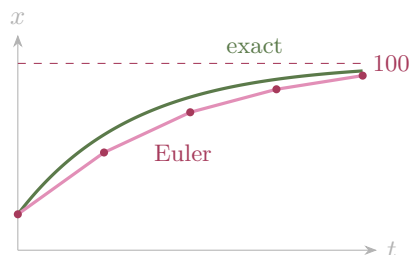
What the picture shows. The arrows are nothing but the sign of f transferred to the axis below. On the left they converge on 1 and diverge from 0; on the right they all point the same way, which no sign of f' could have told you.

Exercise 11 Euler on $\dot{x} = 0.1(100 - x)$; is the numerical value above or below; the stability condition; stability versus accuracy.

What this exercise uses. Euler replaces the curve, over one step, by its **tangent at the left endpoint**: $x_{n+1} = x_n + \Delta t f(t_n, x_n)$. The local error is $O(\Delta t^2)$ and the error accumulated over a fixed interval is $O(\Delta t)$.

Solution.

- (a) Step 1: rate = $0.1(100 - 20) = 8$, so $x_1 = 20 + 8 = 28$. Step 2: rate = $0.1(100 - 28) = 7.2$, so $x_2 = 28 + 7.2 = 35.2$. Exact: $x(2) = 100 - 80e^{-0.2} \approx 100 - 80(0.8187) \approx 34.50$. The scheme overshoots by about 0.7.
- (b) **Above.** x is concave here: $\ddot{x} = -0.1\dot{x} < 0$ since $\dot{x} > 0$. The graph of a concave function lies below each of its tangents, so following the tangent at the left endpoint lands above the curve — at every step.
- (c) By induction: $x_{n+1} = x_n + \Delta t ax_n = (1 + a\Delta t)x_n$, hence $x_n = (1 + a\Delta t)^n x_0$. The sequence stays bounded iff $|1 + a\Delta t| \leq 1$, i.e. for $a < 0$ iff $\Delta t \leq 2/|a| = 2\tau$. With $a = -1/5$: $\Delta t \leq 10$.
- (d) Because the two are different statements. Boundedness only says the iterates do not diverge; the accumulated error is still $O(\Delta t)$, so a stable simulation with a large step can sit far from the true solution — as in (a), where nothing exploded and the answer was nonetheless wrong in the first decimal.



Each segment is a tangent taken at its left endpoint.

What the picture shows. The polygon rides *above* the curve throughout, which is the picture of the concavity argument in (b): every tangent of a concave curve lies above it.

Remark. Halving Δt roughly halves the gap between polygon and curve — that is what “global error $O(\Delta t)$ ” means in practice. And if a simulation ever explodes, suspect Δt before suspecting the model.